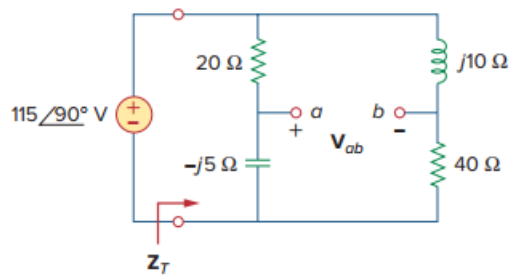


Problem

For the circuit in Fig. 9.73, calculate $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$.



Step-by-step solution

Step 1 of 5

Refer to Figure 9.73 in the text book.

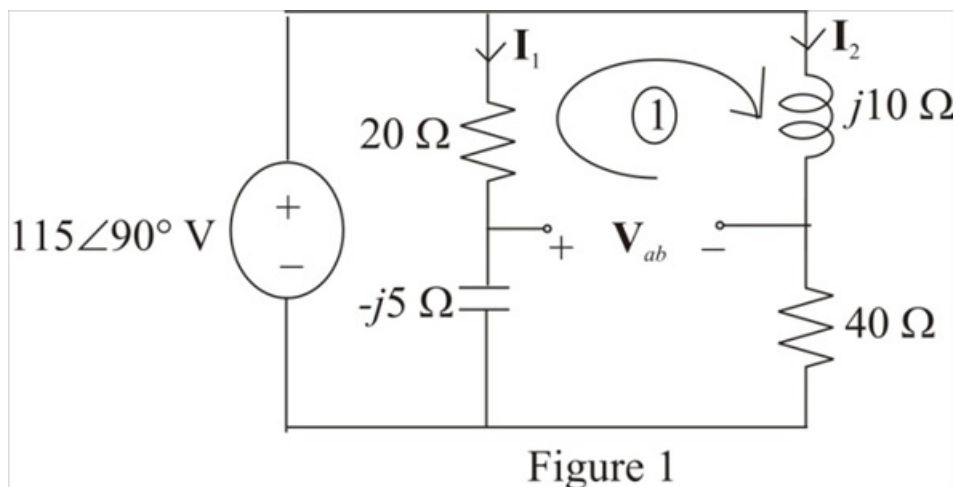
Calculate the value of the impedance $\mathbf{Z}_T$.

$$\begin{aligned}\mathbf{Z}_T &= (20 - j5) \parallel (40 + j10) \\ &= \frac{(20 - j5)(40 + j10)}{(20 - j5) + (40 + j10)} \\ &= \frac{850}{60 + j5} \\ &= 14.07 - j1.17 \, \Omega\end{aligned}$$

Therefore, the value of the impedance $\mathbf{Z}_T$ is, $\boxed{14.07 - j1.17 \, \Omega}$.

Step 2 of 5

Identify the currents in the circuit diagram shown in Figure 1.



Step 3 of 5

Calculate the value of the current $\mathbf{I_1}$.

$$\begin{aligned}\mathbf{I_1} &= \frac{115\angle 90^\circ}{20-j5} \\ &= \frac{115\angle 90^\circ}{20.61\angle -14.04^\circ} \\ &= 5.58\angle 104.04^\circ \text{ A}\end{aligned}$$

Therefore, the value of the current $\mathbf{I_1}$ is, $5.58\angle 104.04^\circ \text{ A}$.

Step 4 of 5

Calculate the value of the current $\mathbf{I_2}$.

$$\begin{aligned}\mathbf{I_2} &= \frac{115\angle 90^\circ}{40+j10} \\ &= \frac{115\angle 90^\circ}{41.23\angle 14.04^\circ} \\ &= 2.79\angle 75.96^\circ \text{ A}\end{aligned}$$

Therefore, the value of the current $\mathbf{I_2}$ is, $2.79\angle 75.96^\circ \text{ A}$.

Step 5 of 5

Apply Kirchhoff's Voltage Law in loop-1.

$$\begin{aligned}-\mathbf{V_{ab}} - \mathbf{I_1}(20) + \mathbf{I_2}(j10) &= 0 \\ \mathbf{V_{ab}} &= \mathbf{I_2}(10\angle 90^\circ) - \mathbf{I_1}(20)\end{aligned}$$

Substitute $2.79\angle 75.96^\circ \text{ A}$ for $\mathbf{I_2}$ and $5.58\angle 104.04^\circ \text{ A}$ for $\mathbf{I_1}$.

$$\begin{aligned}\mathbf{V_{ab}} &= (2.79\angle 75.96^\circ)(10\angle 90^\circ) - (5.58\angle 104.04^\circ)(20) \\ &= 27.9\angle 165.96^\circ - 111.6\angle 104.04^\circ \\ &= 101.5\angle -90^\circ \text{ V}\end{aligned}$$

Therefore, the value of the voltage $\mathbf{V_{ab}}$ is, $101.5\angle -90^\circ \text{ V}$.